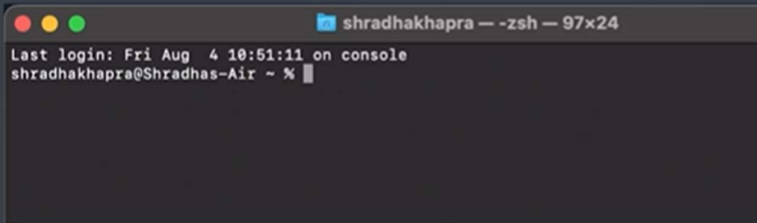


→Terminal

The Terminal

A text input and output environment.



Different Terms

- **Command Line** - any interface that is used by entering textual commands (gen. **Windows** centric)
- **Terminal** - This is a type of command line (gen. **Mac** centric)
- **Console** - A command-line interface used to work with your computer
- **Shell** - A program running on terminal
- **Bash** - A popular shell on Mac OS/ Linux
- **Z-Shell** - Another shell (default)

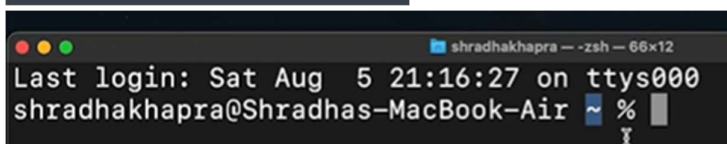
→Installing git bash

Basics

ls = list files
(show my files)

pwd = print working directory
(where am i?)

clear = clear screen



~ this means home directory in mac

→ls

```
shradhakhapra@Shradhas-MacBook-Air ~ % ls
Alpha
AndroidStudioProjects
Apna-College-CppCourse
Applications
C Tutorials
Classroom
DP
Desktop
Documents
Downloads
Extra Code
Graphs
HTML Tutorial
Heaps
IdeaProjects
Library
Movies
Music
Pictures
Postman
Public
PycharmProjects
Python
SegmentTrees
Solution.java
Tries
Utils
WebClassroom
WebDevelopment
bank
encapsulation
homebrew
images
newpackage
prism
shradhakhapra@Shradhas-MacBook-Air ~ %
```

→pwd

```
shradhakhapra@Shradhas-MacBook-Air ~ % pwd
/Users/shradhakhapra
shradhakhapra@Shradhas-MacBook-Air ~ %
```

→clear

```
shradhakhapra@Shradhas-MacBook-Air ~ % ls
Alpha
AndroidStudioProjects
Apna-College-CppCourse
Applications
C Tutorials
Classroom
DP
Desktop
Documents
Downloads
Extra Code
Graphs
HTML Tutorial
Heaps
IdeaProjects
Library
Movies
Music
Pictures
Postman
Public
PycharmProjects
Python
SegmentTrees
Solution.java
Tries
Utils
WebClassroom
WebDevelopment
bank
encapsulation
homebrew
images
newpackage
prism
shradhakhapra@Shradhas-MacBook-Air ~ % clear
```

```
shradhakhapra@Shradhas-MacBook-Air ~ %
```

Navigation

Inside & Outside Directories

cd = change directory

cd Desktop

cd .. = back button

→cd

```
shradhakhapra@Shradhas-MacBook-Air Desktop % pwd
/Users/shradhakhapra/Desktop
shradhakhapra@Shradhas-MacBook-Air Desktop % ls
Delta      Extra Data  coding-ide
shradhakhapra@Shradhas-MacBook-Air Desktop % cd Delta
shradhakhapra@Shradhas-MacBook-Air Delta % pwd
/Users/shradhakhapra/Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % ls
CSS        JS          README.md
shradhakhapra@Shradhas-MacBook-Air Delta % cd CSS
shradhakhapra@Shradhas-MacBook-Air CSS % pwd
/Users/shradhakhapra/Desktop/Delta/CSS
shradhakhapra@Shradhas-MacBook-Air CSS %
```

→back cd ..

```
shradhakhapra@Shradhas-MacBook-Air Delta % pwd
/Users/shradhakhapra/Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % cd ..
shradhakhapra@Shradhas-MacBook-Air Desktop % pwd
/Users/shradhakhapra/Desktop
shradhakhapra@Shradhas-MacBook-Air Desktop % cd ..
shradhakhapra@Shradhas-MacBook-Air ~ % pwd
/Users/shradhakhapra
shradhakhapra@Shradhas-MacBook-Air ~ % cd ..
shradhakhapra@Shradhas-MacBook-Air /Users % pwd
/Users
```

→if we need to move two folders at a time and also we need to find unknown not file

```
shradhakhapra@Shradhas-MacBook-Air ~ % pwd
/Users/shradhakhapra
shradhakhapra@Shradhas-MacBook-Air ~ % cd Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % pwd
/Users/shradhakhapra/Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % cd HTML
cd: no such file or directory: HTML
shradhakhapra@Shradhas-MacBook-Air Delta %
```

→if we need to move two steps back

```
shradhakhapra@Shradhas-MacBook-Air Delta % pwd
/Users/shradhakhapra/Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % cd ../../
shradhakhapra@Shradhas-MacBook-Air ~ % pwd
/Users/shradhakhapra
shradhakhapra@Shradhas-MacBook-Air ~ %
```

Paths

Absolute & Relative

cd Desktop/Delta (relative path)

cd /Users/shradhakhapra/Desktop (absolute path)

/ = root directory

~ = home directory

Paths

Absolute & Relative

`cd Desktop/Delta` (relative path)

→ current location

`cd /Users/shradhakhapra/Desktop` (absolute path)

`/` = root directory

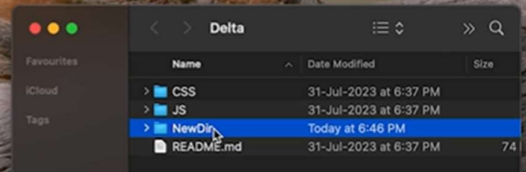
`~` = home directory

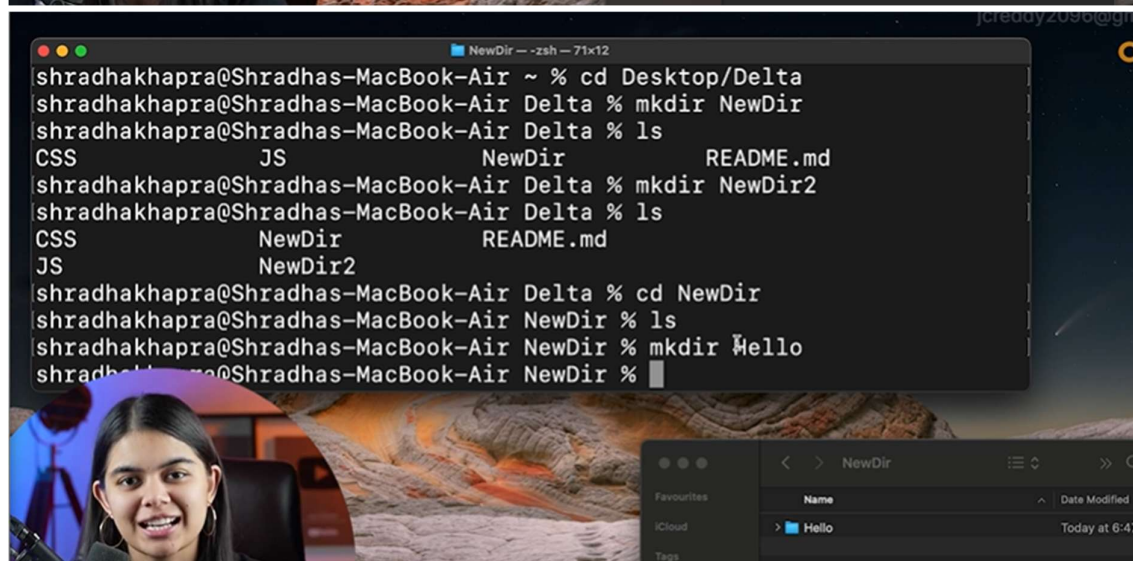
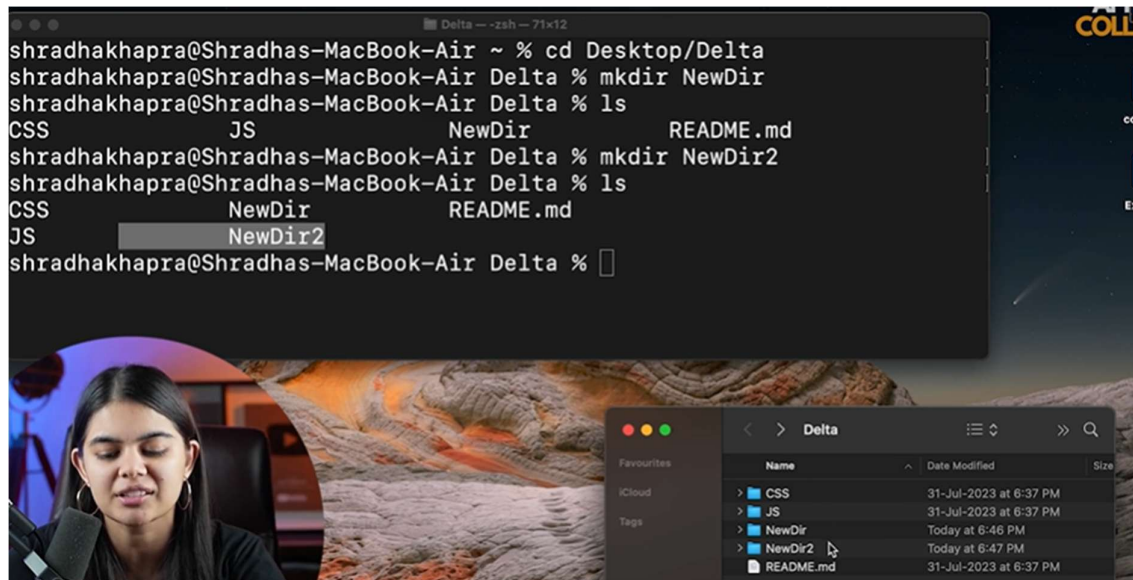
```
shradhakhapra@Shradhas-MacBook-Air ~ % pwd
/Users/shradhakhapra
shradhakhapra@Shradhas-MacBook-Air ~ % cd /Users/shradhakhapra/Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % cd ../../
shradhakhapra@Shradhas-MacBook-Air ~ % cd Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta %
```

Making Directories

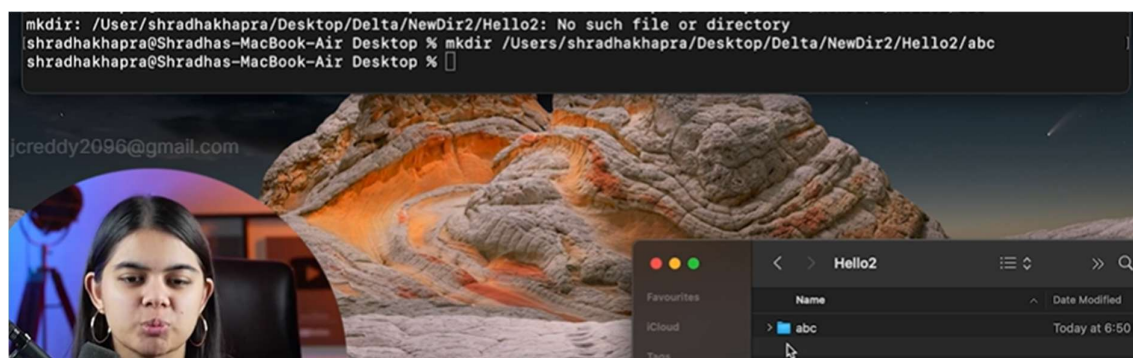
mkdir = make directory
folders

```
shradhakhapra@Shradhas-MacBook-Air ~ % cd Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % mkdir NewDir
shradhakhapra@Shradhas-MacBook-Air Delta %
```





→ Absolute path



Flags

Flags are characters that we pass with commands to modify their behaviour

manual Command

man ls - give info about ls command

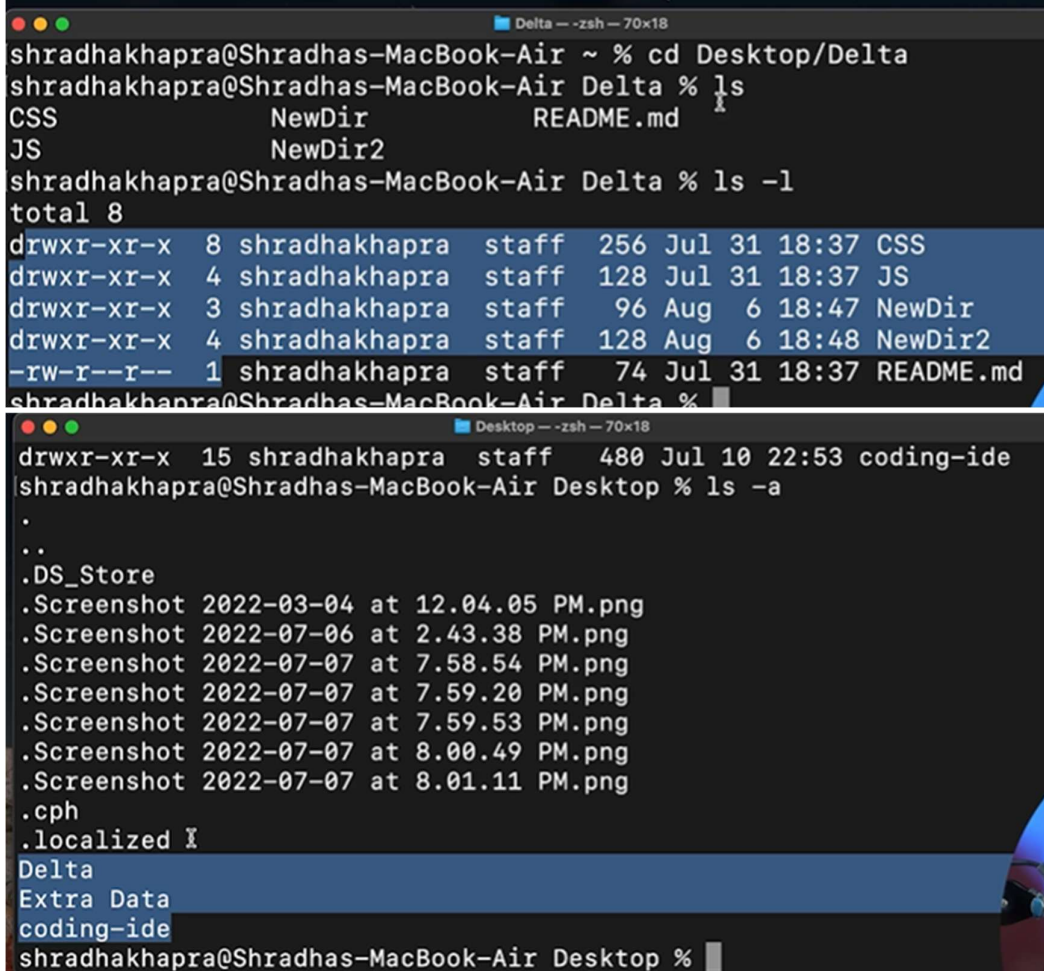
man mkdir - give info about mkdir command

with Flags

`ls -l`

`ls -a`

`ls -la`



The first screenshot shows a terminal window titled "Delta -- zsh -- 70x18". The user is in the directory ~/Desktop/Delta. They run `ls`, which lists files: CSS, NewDir, README.md, JS, and NewDir2. Then they run `ls -l`, which shows a detailed listing of the same files with permissions, sizes, and timestamps.

```
shradhakhapra@Shradhas-MacBook-Air ~ % cd Desktop/Delta
shradhakhapra@Shradhas-MacBook-Air Delta % ls
CSS          NewDir      README.md
JS           NewDir2
shradhakhapra@Shradhas-MacBook-Air Delta % ls -l
total 8
drwxr-xr-x  8 shradhakhapra  staff   256 Jul 31 18:37 CSS
drwxr-xr-x  4 shradhakhapra  staff   128 Jul 31 18:37 JS
drwxr-xr-x  3 shradhakhapra  staff    96 Aug  6 18:47 NewDir
drwxr-xr-x  4 shradhakhapra  staff   128 Aug  6 18:48 NewDir2
-rw-r--r--  1 shradhakhapra  staff    74 Jul 31 18:37 README.md
shradhakhapra@Shradhas-MacBook-Air Delta %
```

The second screenshot shows a terminal window titled "Desktop -- zsh -- 70x18". The user is in the Desktop directory. They run `ls -a`, which lists all files and directories, including hidden ones like .DS_Store, .cph, and .localized. The output shows a list of screenshot files and other system files.

```
drwxr-xr-x 15 shradhakhapra  staff   480 Jul 10 22:53 coding-ide
shradhakhapra@Shradhas-MacBook-Air Desktop % ls -a
.
..
.DS_Store
.Screenshot 2022-03-04 at 12.04.05 PM.png
.Screenshot 2022-07-06 at 2.43.38 PM.png
.Screenshot 2022-07-07 at 7.58.54 PM.png
.Screenshot 2022-07-07 at 7.59.20 PM.png
.Screenshot 2022-07-07 at 7.59.53 PM.png
.Screenshot 2022-07-07 at 8.00.49 PM.png
.Screenshot 2022-07-07 at 8.01.11 PM.png
.cph
.localized
Delta
Extra Data
coding-ide
shradhakhapra@Shradhas-MacBook-Air Desktop %
```

```
coding-ide  
shradhakhapra@Shradhas-MacBook-Air Desktop % ls -la
```

```
Desktop -- zsh -- 70x18  
-rw-r--r--@ 1 shradhakhapra staff 0 Jul 6 2022 .Screenshot 20  
22-07-06 at 2.43.38 PM.png  
-rw-r--r--@ 1 shradhakhapra staff 0 Jul 7 2022 .Screenshot 20  
22-07-07 at 7.58.54 PM.png  
-rw-r--r--@ 1 shradhakhapra staff 0 Jul 7 2022 .Screenshot 20  
22-07-07 at 7.59.20 PM.png  
-rw-r--r--@ 1 shradhakhapra staff 0 Jul 7 2022 .Screenshot 20  
22-07-07 at 7.59.53 PM.png  
-rw-r--r--@ 1 shradhakhapra staff 0 Jul 7 2022 .Screenshot 20  
22-07-07 at 8.00.49 PM.png  
-rw-r--r--@ 1 shradhakhapra staff 0 Jul 7 2022 .Screenshot 20  
22-07-07 at 8.01.11 PM.png  
drwxr-xr-x 3 shradhakhapra staff 96 Jun 11 2021 .cph  
-rw-r--r-- 1 shradhakhapra staff 0 Mar 24 2021 .localized  
drwxr-xr-x 10 shradhakhapra staff 320 Aug 6 18:47 Delta  
drwxr-xr-x 33 shradhakhapra staff 1056 Aug 6 19:03 Extra Data  
drwxr-xr-x 15 shradhakhapra staff 480 Jul 10 22:53 coding-ide  
shradhakhapra@Shradhas-MacBook-Air Desktop %
```

→ Touch command

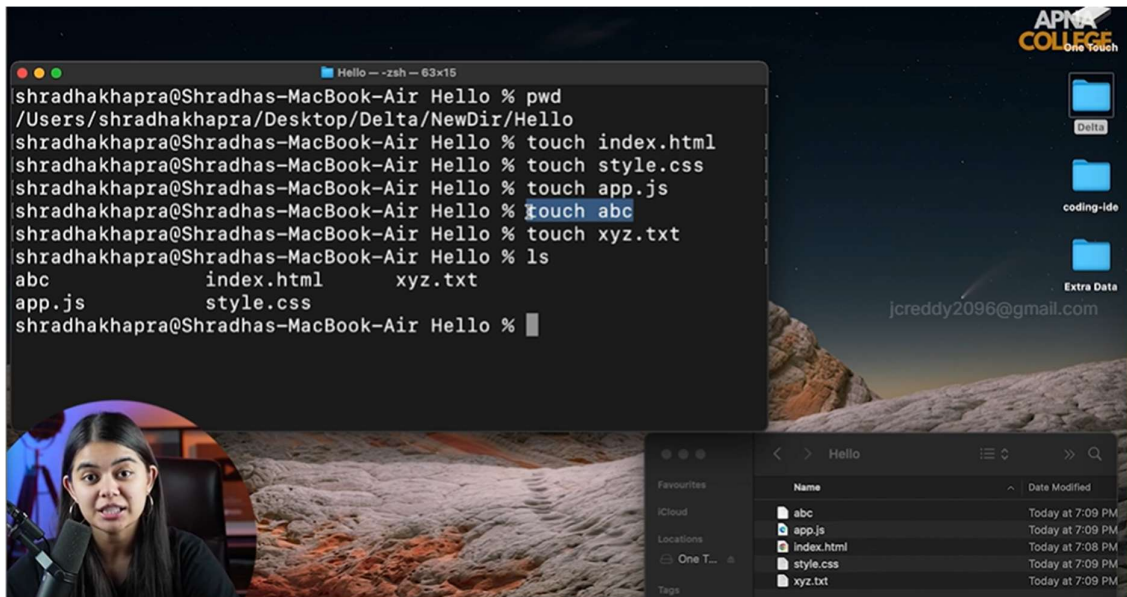
Touch Command

Used to create files

```
touch index.html
```

```
touch app.js
```

```
touch abc.txt
```

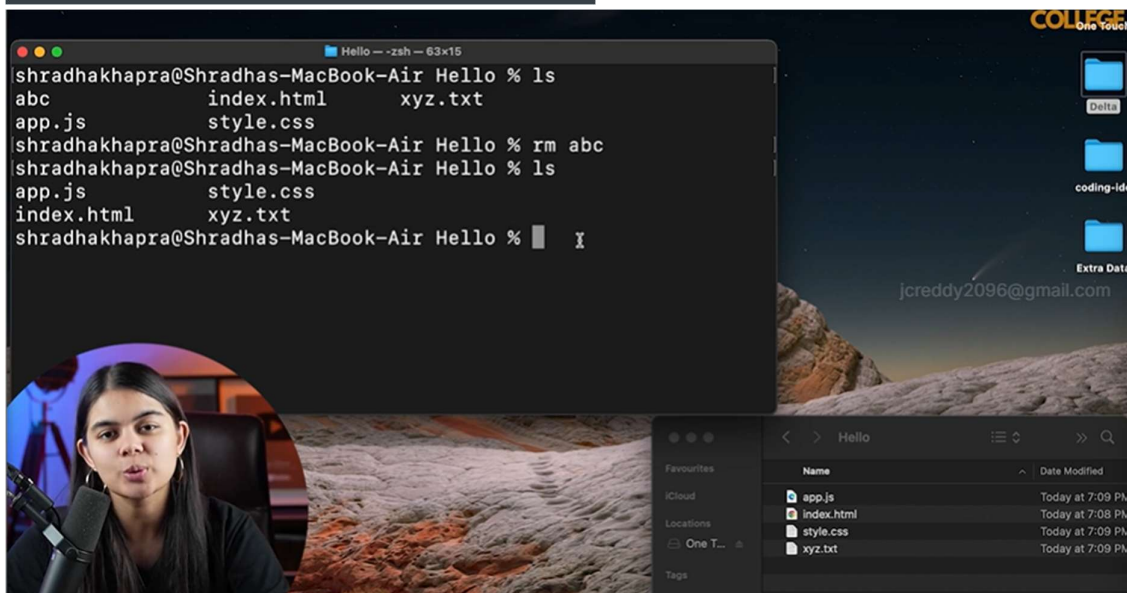



Deleting Files & Folders

rm - removes files

rmdir - removes empty folders

rm -rf - removes any folders



delete → sensitive

Deleting Files & Folders

rm - removes files

rmdir - removes empty folders

rm -rf - removes any folders

recursive delete

forced

recursive force
[remove with force]